

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 1

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 2 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

Dr Draga Pihler-Puzovic (Year Tutor)

draga.pihler-puzovic@manchester.ac.uk

PHYS10302 Vibrations and waves

Week 1 Questions

- A mass of 0.200 kg is attached to a spring with a spring constant of 720 N/m. What is the period?
 - A mass of 1.20 kg is attached to a spring. It vibrates with a frequency of 10.2 Hz. What is the spring constant?
 - A mass is attached to a spring, and the system has angular frequency $\omega = 24 \text{ rad/s}$. At time $t = 0$ the mass is at its equilibrium position ($x = 0$) when it is given a velocity $v = 6.0 \text{ m/s}$. Calculate the displacement of the mass at time $t = 5.0 \text{ s}$.
 - A pendulum has a 200 g mass on a 60.0 cm string. What are the angular frequency ω and the frequency f ?
 - How long should a pendulum be to have a period of 1.0 s?

- Show that we can write the general solution of an undamped oscillator,

$$x(t) = A \sin(\omega t + \phi),$$

as the sum of a sine and a cosine term. What is the relationship between the two sets of constants?

- Consider a diatomic molecule A^+ bound to B^- . A suitable formula for the mutual potential energy of the two ions is

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r} + \frac{B}{r^9}$$

What is the role of the second term and for what region in r does it dominate?

Find an expression for the positive constant B valid at the equilibrium separation length $r = R$.

At this equilibrium separation $r = R$, what is the value of the spring constant k associated with the small amplitude simple harmonic oscillations of the system? (hint : consider a series expansion of the potential and you may assume the distance from the equilibrium separation is small).

- The general form of the total energy E of any system undergoing SHM is given by

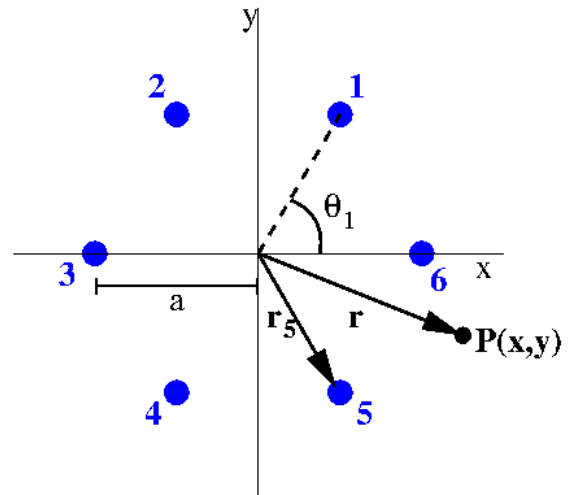
$$E = \frac{1}{2}\alpha v^2 + \frac{1}{2}\beta x^2$$

where α and β are positive constants with units of mass and N m^{-1} respectively. By differentiating this expression with respect to time and assuming the total energy is

constant, find an expression for the angular frequency of this general system. Hence, by using your knowledge of the angular frequency of a simple pendulum and taking $\alpha = m$, write down an expression for the total energy of a simple pendulum.

E&M Tutorial Sheet 1: Coulomb's Law and Superposition

- 1 (Revision) Write down a vector calculus expression for the work done, W , by a varying force $\mathbf{F}(\mathbf{x})$ when applied to an object moving along a curve C .
- 2 Two positive charges, $+q$, and one negative charge, $-q$, are placed at the vertexes of an equilateral triangle with sides of length, a . What is the force on the negative charge? *Remember to give the magnitude and direction. Hint: if you set up your model correctly you can use symmetry to simplify this question.*
- 3 Three identical charges are placed in a row, with a distance a between each.
 - (a) If the middle charge is displaced by a small amount along the line, what is the restoring force?
Hint: Write the separation of the charges in terms of the equilibrium separation, a , and the displacement of the middle charge, x . Use Coulomb's law and superposition, thinking carefully about directions. Use the Binomial expansion to write the force out in a series, so that for small displacements you can neglect the higher terms.
 - (b) What happens if the middle charge has the opposite sign to the other two?
 - (c) Without detailed calculation, how would your answers to (a) and (b) change if the displacement was perpendicular to the line?
- 4 Six identical charges, q , are fixed at the vertices of a regular hexagon in the xy plane, centred on the origin, with sides of length a as shown on the diagram below.
 - (a) Write down the Cartesian position vector, $\mathbf{r}_n$, of the n^{th} charge. *Hint: first find the angle θ_n for the n^{th} charge from the origin.*
 - (b) Find the electric field at any point $P(x,y)$ with position vector $\mathbf{r}$ in the xy plane. *Hint: find the vector between $\mathbf{r}$ and $\mathbf{r}_n$ and then use superposition.*
 - (c) What is the electric field on the x axis? *You should be able to use symmetry to predict its y component, but also check that you can see it from the equations.*
 - (d) What is the field at the origin? *Again, use symmetry and check that the equations agree.*
 - (e) What would be the field on the x axis a long distance away from the origin? *Once again, use your intuition and check that the equations agree.*
 - (f) If one of the six charges were removed, what would the field be at the origin? *Hint: there's an easy and a hard way to do this last part. The easy way is to use superposition: removing a positive charge is identical to adding a negative charge at the same point.*
- 5 A line charge of uniform charge density λ extends from the origin along the x -axis towards infinity. Calculate the electric field at the point $(-a,0,0)$. *Hint: calculate the small electric field from the element of charge between $x \rightarrow x + dx$ and then use superposition.*



PHYS10372 Mathematics 2: Tutorial-01 Questions

These questions cover the material from week-1 of PHYS10372: scalar/vector fields and use of the grad (del) ∇ operator.

1. Determine what geometric shapes describe the level surfaces of the scalar field:

$$\psi = \sin^{-1} \left(\frac{z}{\sqrt{x^2 + y^2}} \right).$$

(Hint: it might help to think about your answer in terms of cylindrical polar coordinates.)

2. Find the grad of the following functions:

(a) $\psi = x^2 + y + z^2$

(b) $\psi = e^{(x-y)/2z} + x$

(c) $\psi = 3xyz - x(y^2 - e^z)$

(d) $\psi = \sin(xyz)$

3. Find a unit vector at the point $(1, 1, 1)$ which is normal to the following level surfaces:

(a) $x^2 + y + z^2 = 3$

(b) $e^{(x-y)/2z} + x = 2$

4. Find the rate of increase of the following scalar fields at the point $(1, 1, 1)$ and in the direction between $(1, 1, 1)$ and $(1, 2, 2)$:

(a) $\psi = 3xyz - x(y^2 - e^z)$

(b) $\psi = \sin(xyz)$

[Answers: (a) $\frac{4+e}{\sqrt{2}}$, (b) $\sqrt{2} \cos(1)$]